trouble with having fundamental laws of motion with no inbuilt distinction between past and future, laws whose mathematics treats the future and past of any given moment in exactly the same way.[12]

Rest assured that we will shortly find a way out of the strange place to which an egalitarian use of entropic reasoning has taken us; I'm not going to try to convince you that your memories and records are of a past that never happened (apologies to fans of *The Matrix*). But we will find it very useful to pinpoint precisely the disjuncture between intuition and the mathematical laws. So let's keep following the trail.

A Quagmire

Your intuition balks at a past with higher entropy because, when viewed in the usual forward-time unfolding of events, it would require a spontaneous rise in order: water molecules spontaneously cooling to 0 degrees Celsius and turning into ice, brains spontaneously acquiring memories of things that didn't happen, video cameras spontaneously producing images of things that never were, and so on, all of which seem extraordinarily unlikely—a proposed explanation of the past at which even Oliver Stone would scoff. On this point, the physical laws and the mathematics of entropy agree with your intuition completely. Such a sequence of events, when viewed in the forward time direction from 10 p.m. to 10:30 p.m., goes against the grain of the second law of thermodynamics—it results in a decrease in entropy—and so, although not impossible, it *is* very unlikely.

By contrast, your intuition and experience tell you that a far more likely sequence of events is that ice cubes that were fully formed at 10 p.m. partially melted into what you see in your glass, right now, at 10:30 p.m. But on this point, the physical laws and mathematics of entropy only partly agree with your expectation. Math and intuition concur that *if* there really were fully formed ice cubes at 10 p.m., then the most likely sequence of events would be for them to melt into the partial cubes you see at 10:30 p.m.: the resulting increase in entropy is in line both with the second law of thermodynamics and with experience. But where math and intuition deviate is that our intuition, unlike the math, fails to take account of the likelihood, or lack thereof, of actually having fully formed